

Repetition „Memory“

- What is the main task of a memory?
- Which basic reasons for remembering do you know?
- How would you implement forgetting?
- What is a name?
- What are names used for?
- Why are names so important for remembering?
- How is the memory related to time logic and space logic?

Strong Artificial Intelligence

I Mind

Thought Centre

Prof. Dr. Dr. h. c. Robert Grebner

Technical University of Applied Sciences, Würzburg, 2026-06-12



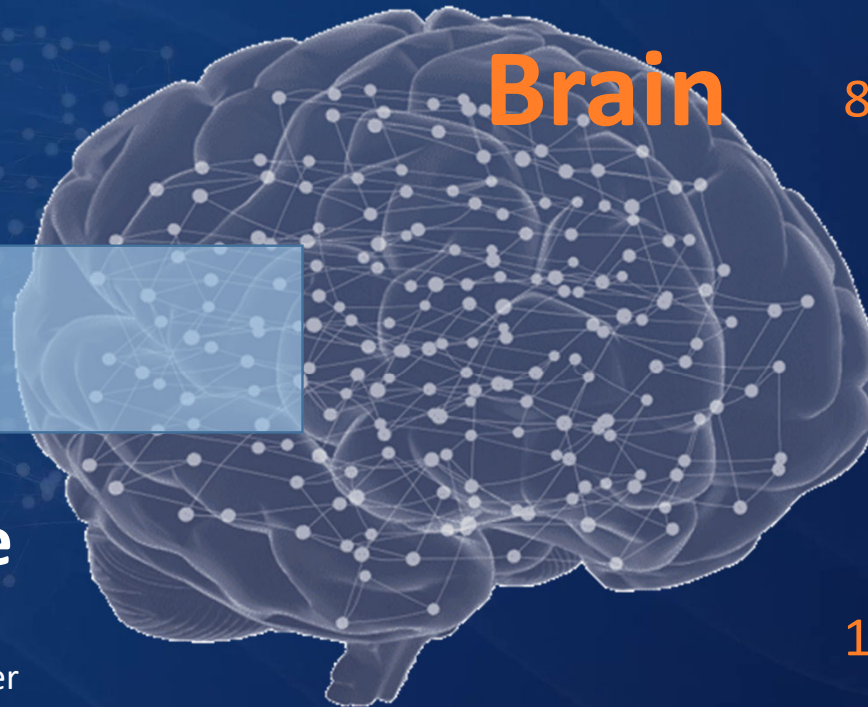
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Brain

86 Billion Neurons

100 Trillion Connections

> 2 Million years evolution

35,000 to 100,000 years old

1.3 kg +/-

20 Watt

Workshop „Mind“

- What is the a mind?
- What is the main task of a mind?
- What are tasks of the mind in respect to our daily life?
- What has the mind to do with writing?
- What is writing?
- What do we think about, when we are writing?
- What is a thought?
- What is the difference between thought and information?

I Mind

- | | |
|------------------|------------------------|
| 1. Thought | Mental Substance |
| 2. Understanding | Associating |
| 3. Reasoning | Decider |
| 4. Learning | Growth |
| 5. Logic | Right or Wrong |
| 6. Initiating | Motion Start-up Setter |
| 7. Conclusion | Processing Core |



1 Thought

Mental Substance

1. Thought

What is a Thought?

Aware Information Just
Processing.

Where are Thoughts?

Mind's eye.

What is the Smallest Possible
Thought?

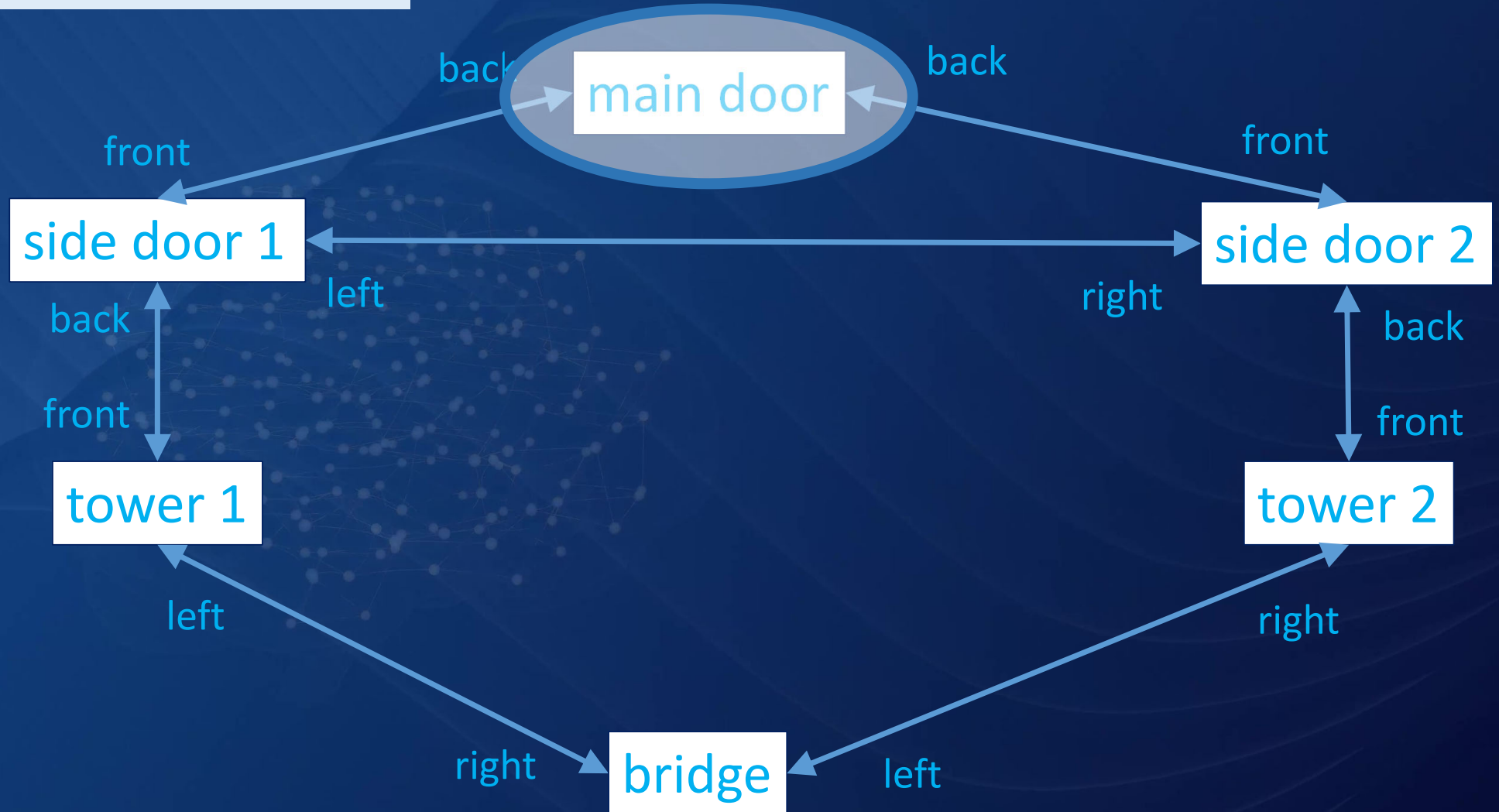
One Entity.

How Many Thoughts Can We
Have in Parallel?

?

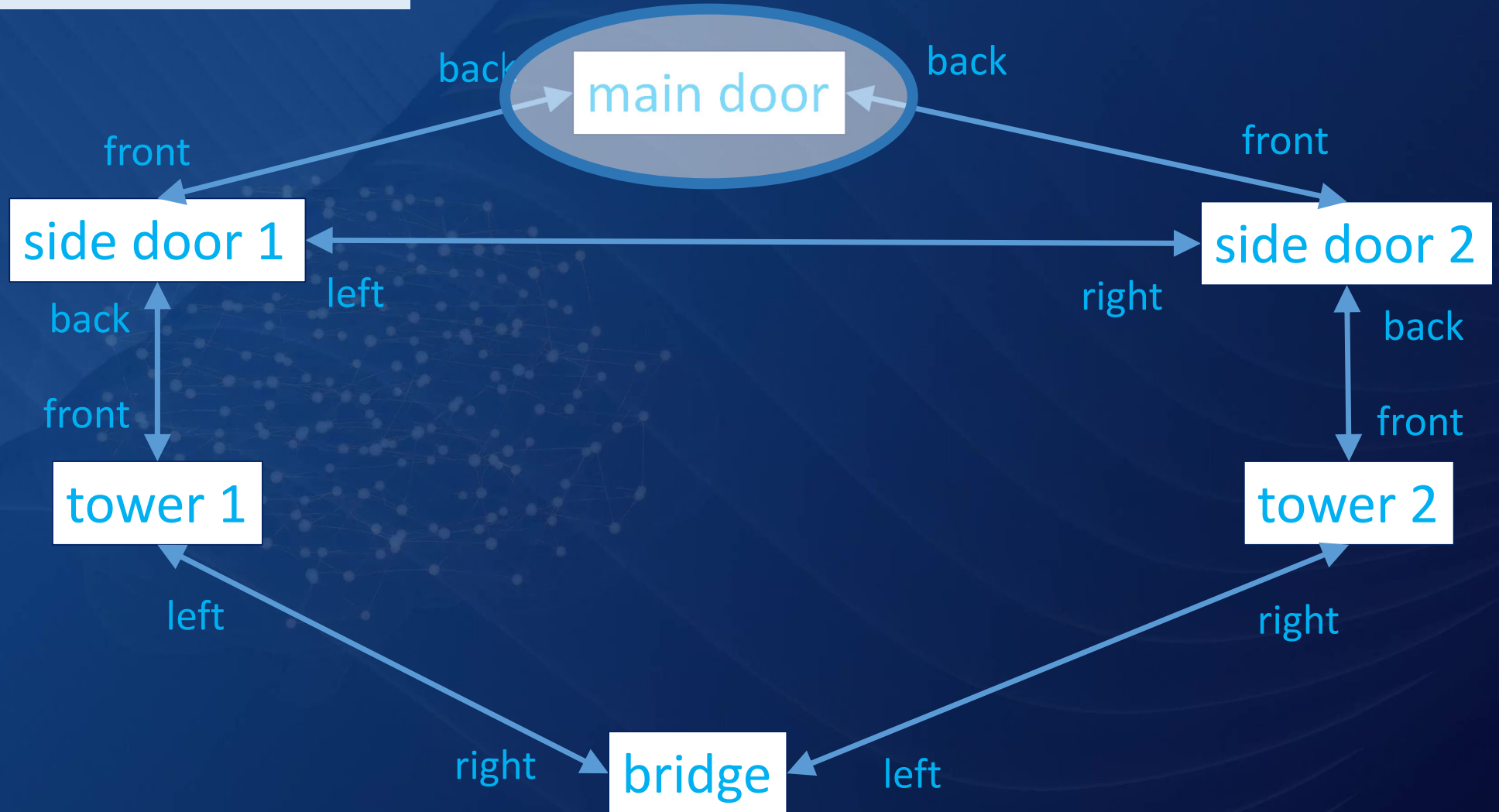
1. Thought

Highlighted Information



1. Thought

Highlighted Information





2 Understanding

Associating

Workshop „Understanding“

- What means understanding?
- Give an example where you understood a topic the first time in your live.
- Give an example of an not understood topic.
- If you understand a phenomenon, what can you use that for?
- How is knoledge and understanding connected?

2. Understanding

What is Understanding?

Taking
Description/Explanation to
Form Consistent Concepts.

What is the Result of
Understanding?

Knowledge.

What Information is Needed?

Information Related to the
Concept that is not
Understood.

What is Changed by
Understanding?

The Information Network of
the Memory.

2. Understanding

Example Time Machine

What is a Time Machine?

Moving our Brain and/or
Body in the Time.

What Restrictions do Time
Machines have?

...

How can you Build a Time
Machine?

...

Why are Time Machines not
Build yet?

High Complexity!
Lack of Technology!



3 Reasoning

Decider

Workshop „Reasoning“

- What is reasoning?
- Give one example of reasoning.
- What is the precondition to make reasoning possible?

3. Reasoning

What is Reasoning?

Gaining new
Insights/Concepts out of
Known Concepts.

What is it Used for Externally?

Decision Making, Action
Plans.

What is it Used for Internally?

Learning.

Precondition?

Logic.



4 Learning

Growth

4. Learning

What is Learning?

Enlarging the Knowledge Base (Known Concepts).

Where Comes the Information From?

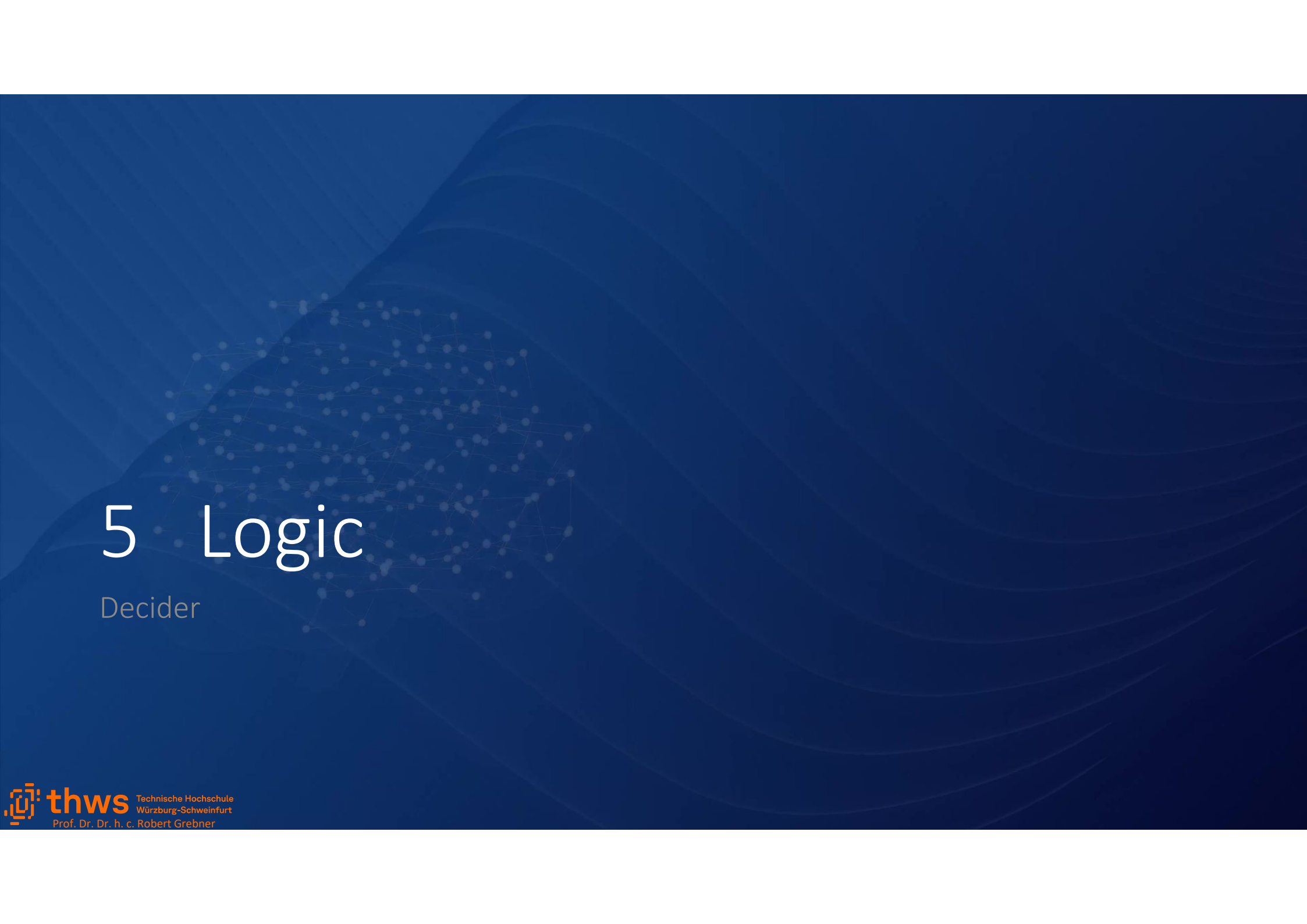
Self Experience or Others Experience.

How is it Done?

Understanding and Reasoning.

Where is the Learned Stored?

(Subjective) Memory.



5 Logic

Decider

Workshop „Logic“

- What is logic?
- What is it used for?
- Where does it come from? (Or how do we learn logic?)

5. Logic

Basic Logics:

- Naming (for information recalls, efficiency)
- Placeholder (variable for information gap)
- Abstraction (for information gap and inference)
- Transitivity (for information inference)
- Exclusivity (for information inference)

Complex Logics:

- Temporal logic (transitivity)
- Spatial logic (part of)
- Concepts, Action plans, ...

Hypothetical Syllogism by Chrysippos (estm. 280-210 B. C.). He

- made a differentiation among object (entity), meaning (concept) and name
- declared statments as true or false
- made a differentiation among propositional, interrogative, request/command and other non propositional sentences
- used fundamental logic units „and“, „not“, „if“, „else“
- created the two-valued axiomatic propositional logic
 - If a causes b then a implies b
 - If a causes b then not b implies not a
 - If a and b not at the same time then a implies not b and b implies not a
 - If a or b then a implies not b
 - If a or b then not a implies b

Transitivity: If a causes b and b causes c then a causes c

5. Logic

Concepts of Time Logic?

before, after, overlapping,
intersecting, ...

Concepts of Number Logic?

greater than, less than, ...

Concepts of Space Logic?

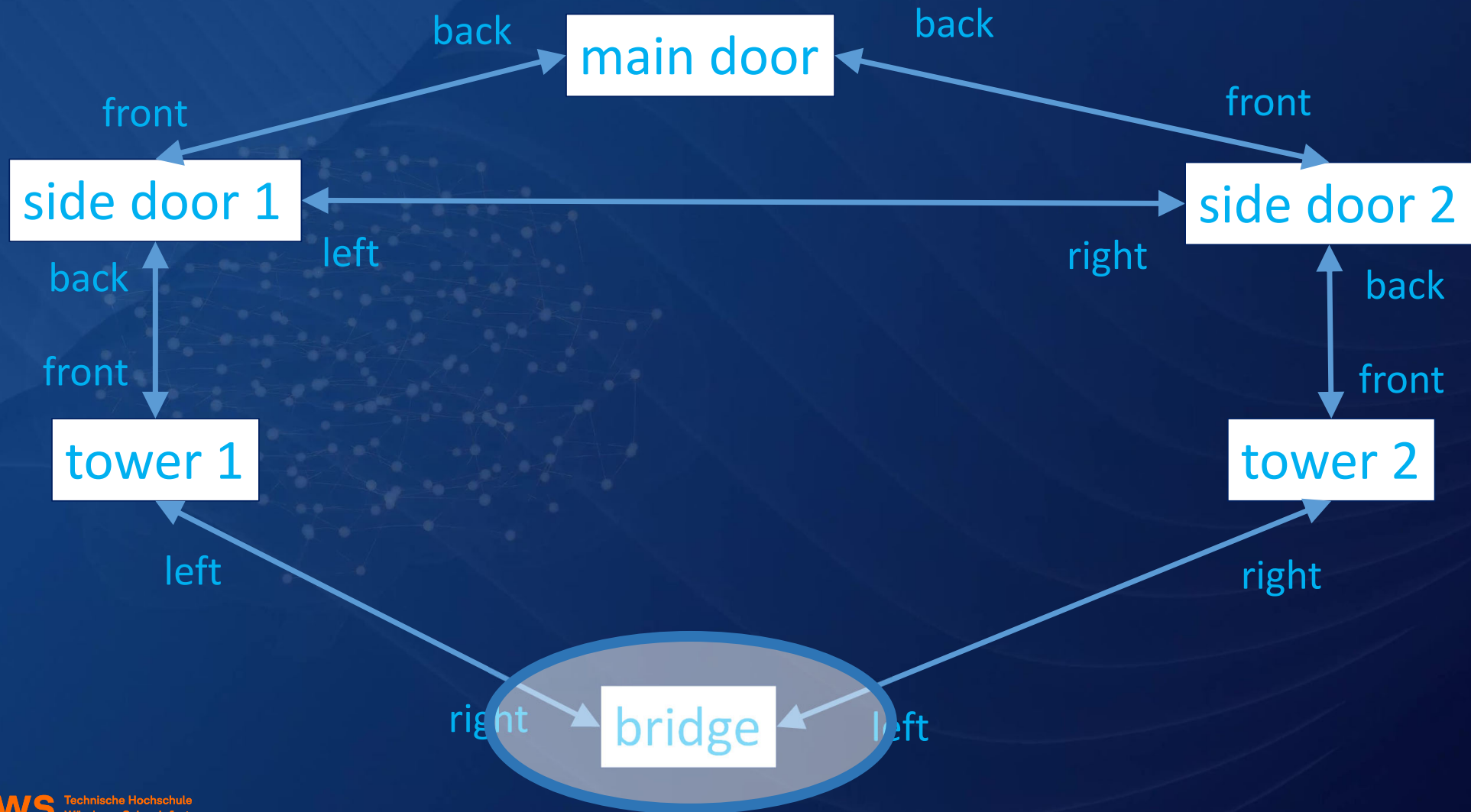
is part of, left, right, front,
back, up, down, overlapping,
distance, is located of, ...

Concepts of Value/Legal/Propositional Logic?

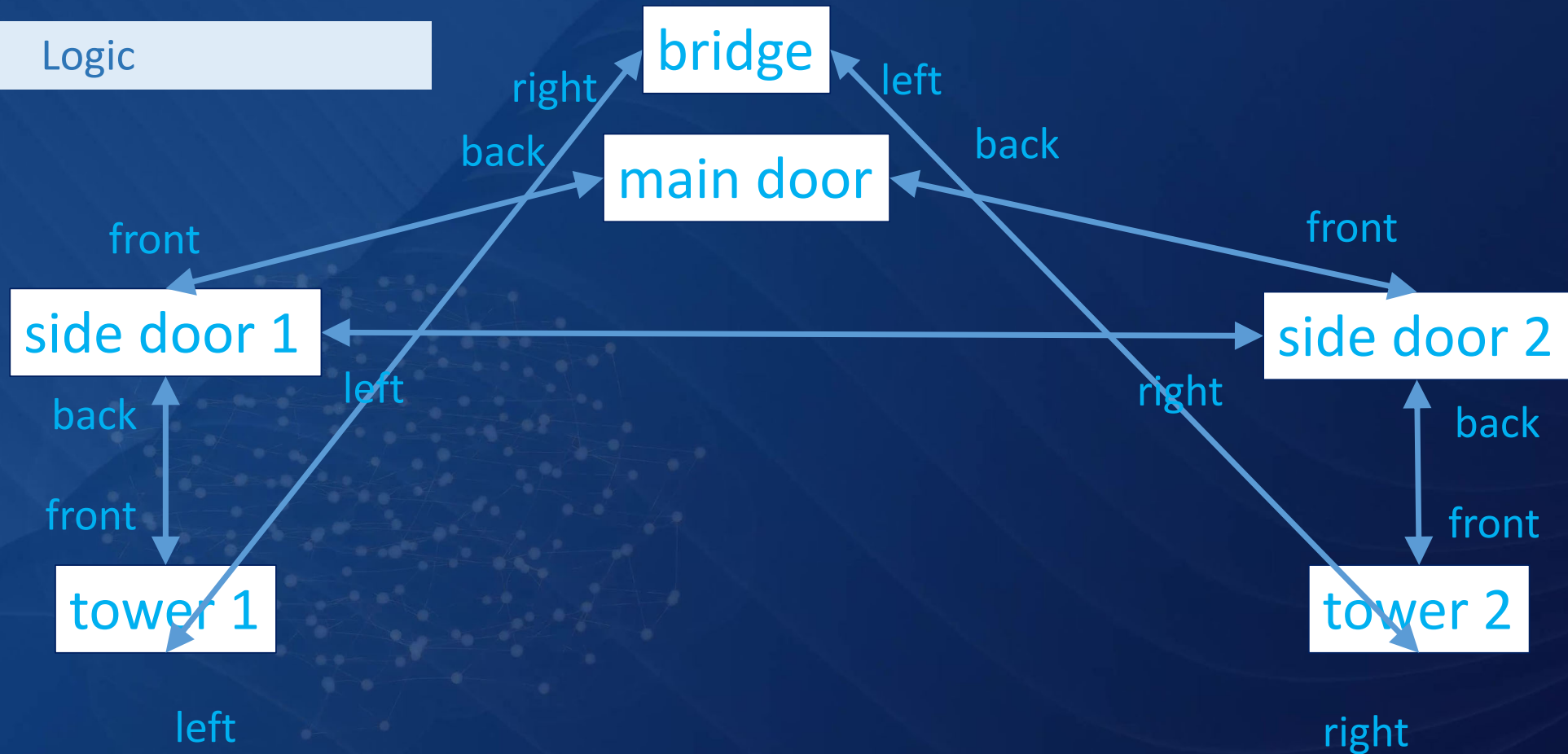
Hypothetical Syllogism, ...

5. Logic

Example: Missing Information

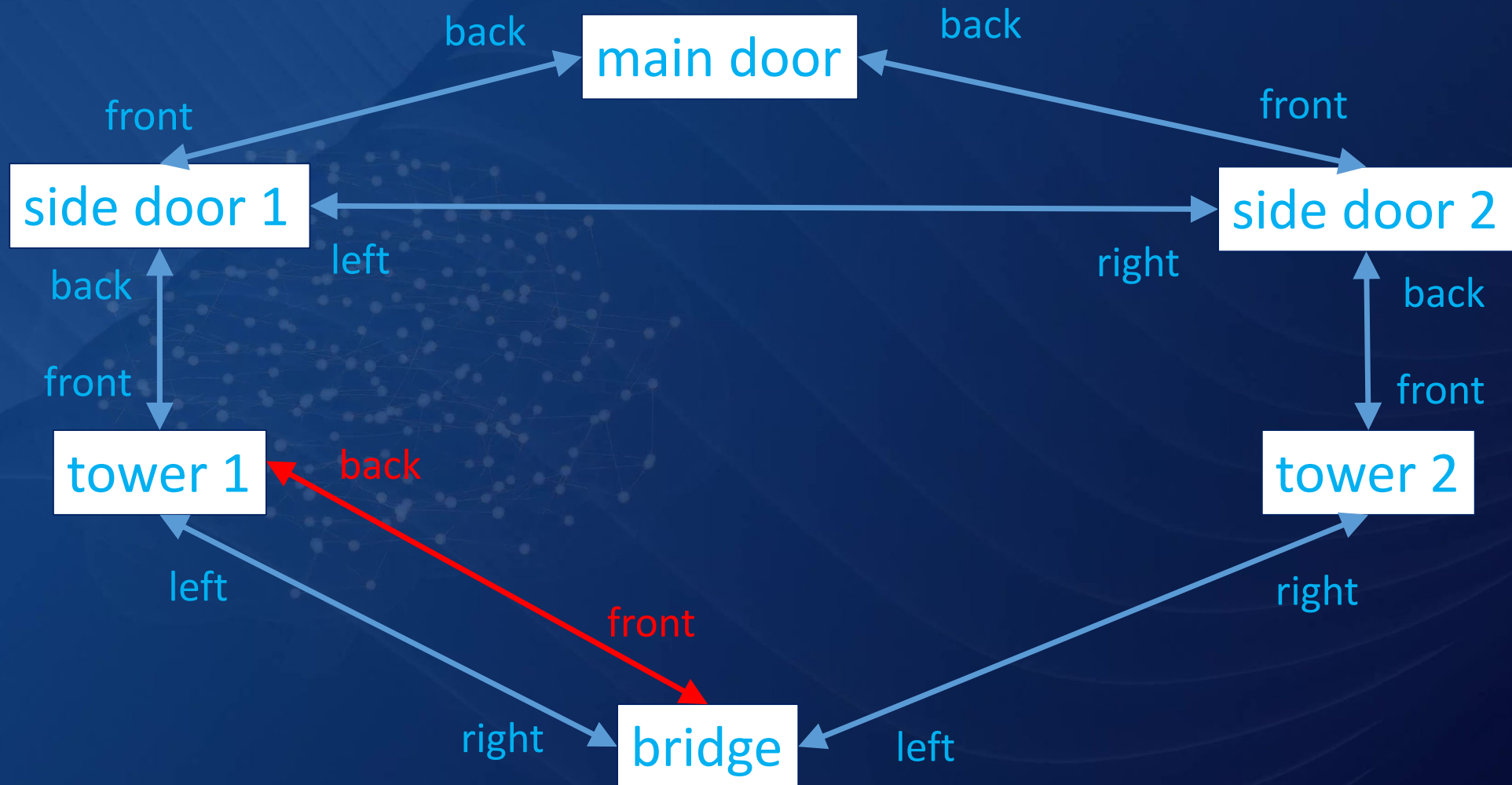


5. Logic



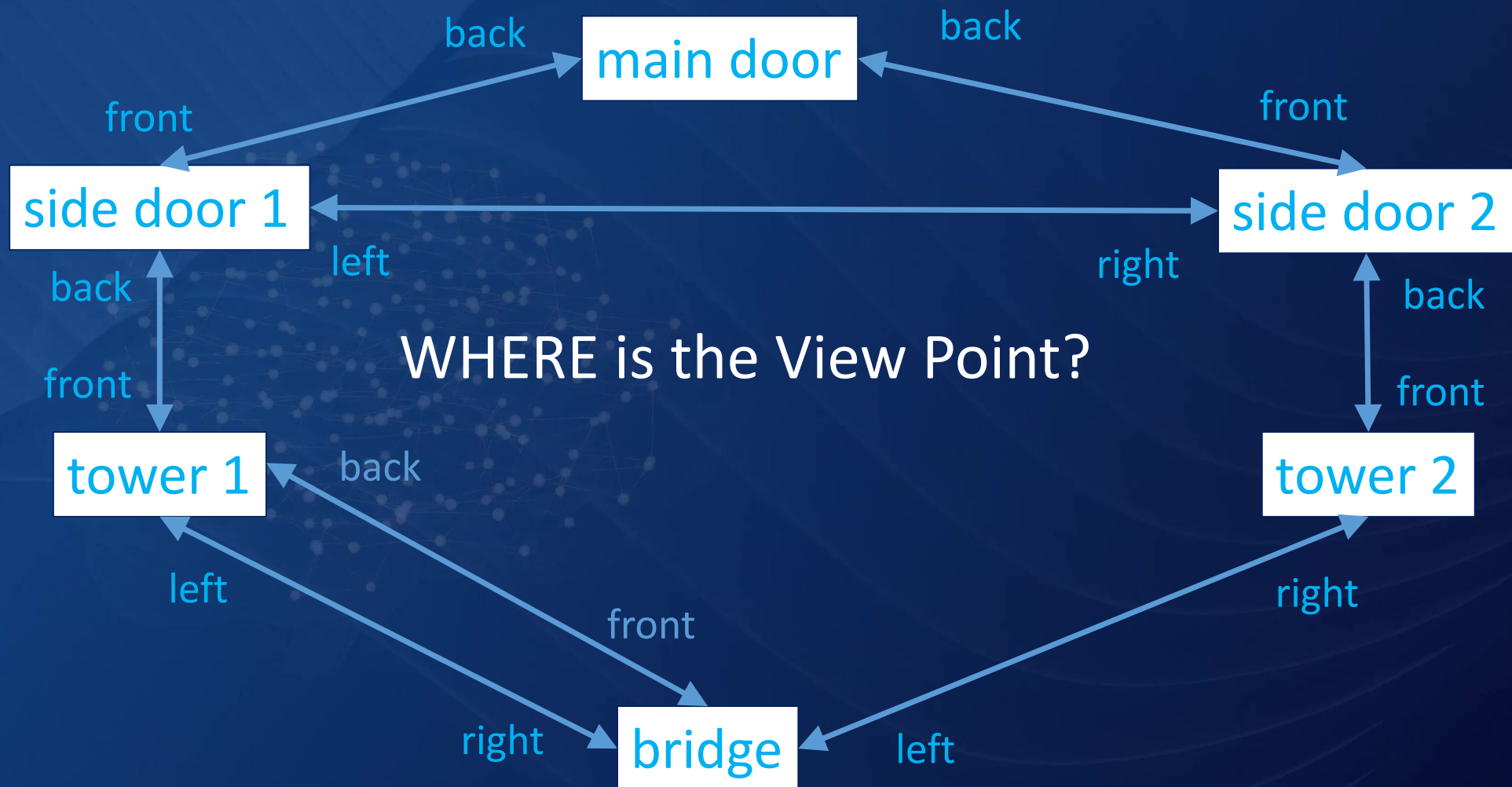
5. Logic

Example: Missing Information



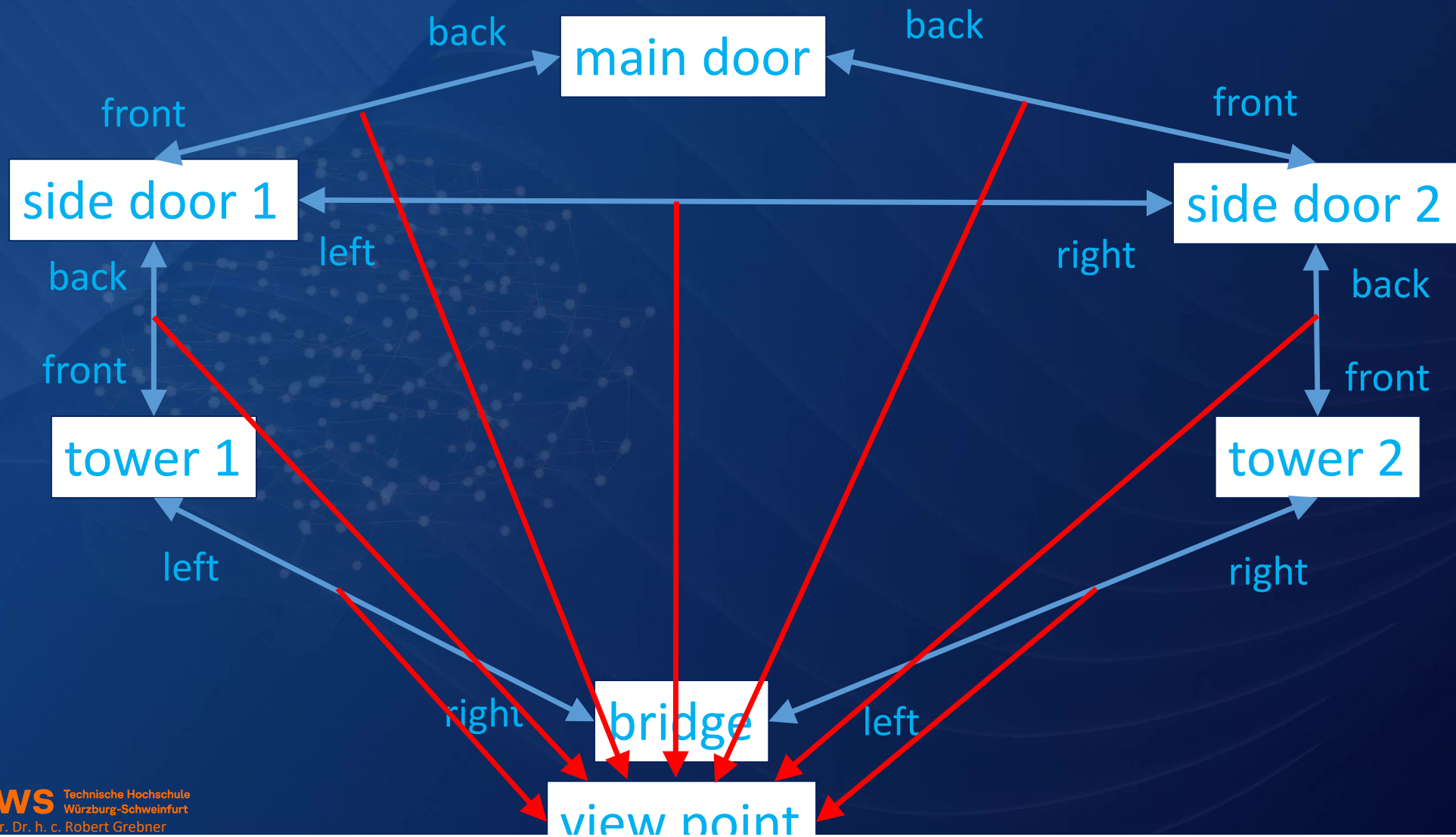
5. Logic

Example: Missing Information



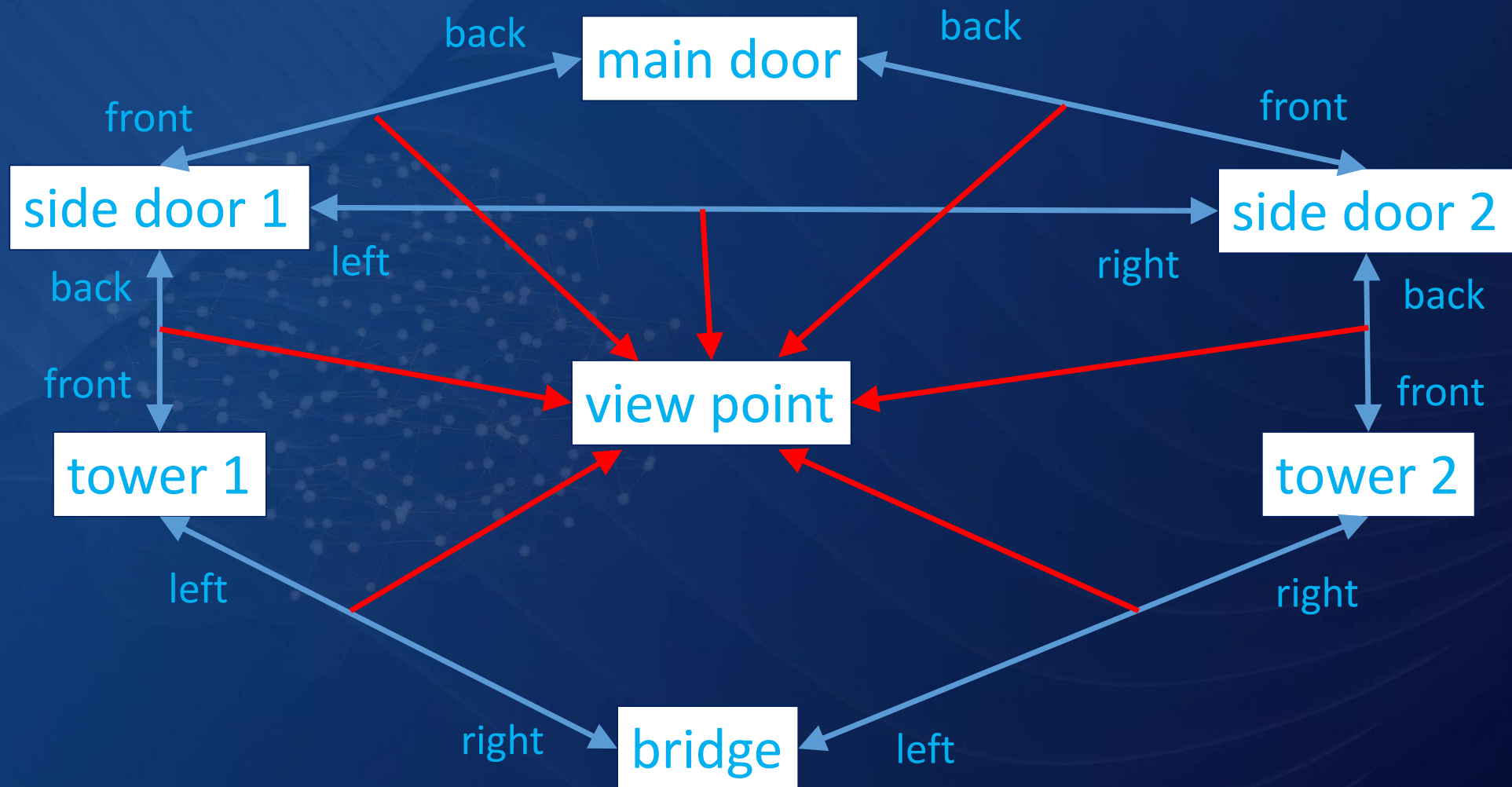
5. Logic

Example: Missing View Point



5. Logic

Example: Castle



Spatial exclusivity fundamentally teaches us:

- I never saw one entity at different places at the same time
- I cannot place myself where another entity already occupies space

Questions:

- Where do we know from, that it is the same time?
- Where do we know from, that it is the same entity?
- How can we explain, if we see the same item at two places?
- What would be, if an entity could be at two places at one time?
- Do we use exclusivity logic also in other areas beyond spatial issues?

Logic included?

exclusivity

Logics excluded?

temporality, spatiality

$$y = f(x) = x \text{ times } x$$

Logic included?

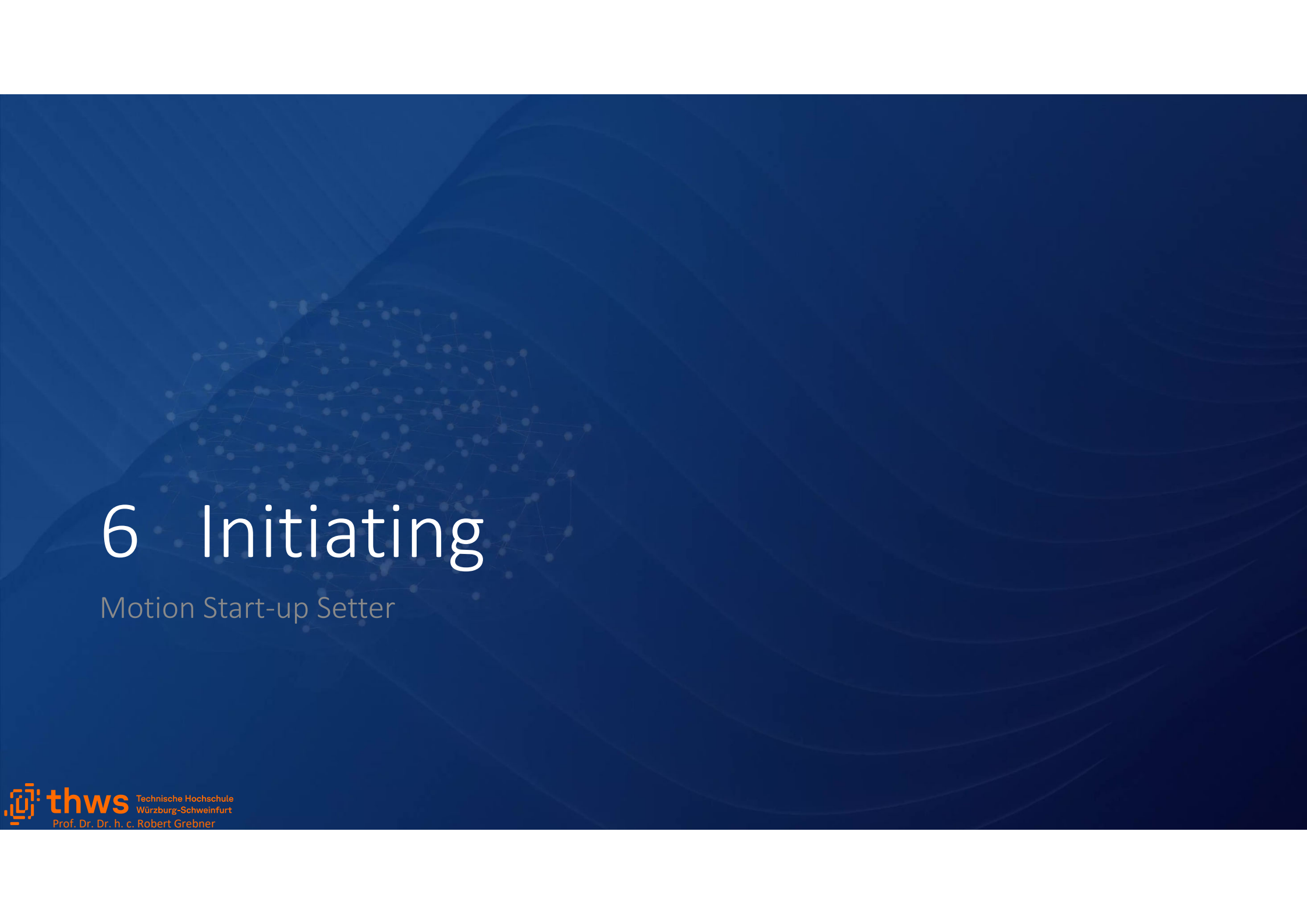
exclusivity, temporality

Logics excluded?

spatiality?

Gulf of Mexico
? - 2025-01-19

Gulf of America
2025-01-20 - ?



6 Initiating

Motion Start-up Setter

6. Initiating

What is Initiated?

Motion, Speech and Thinking Acts.

What Helps?

Action Plans.

Why is Something Initiated?

Impressions are Hurting our Values.

Challenge?

Immediate and Intermediate Feedback by Senses.

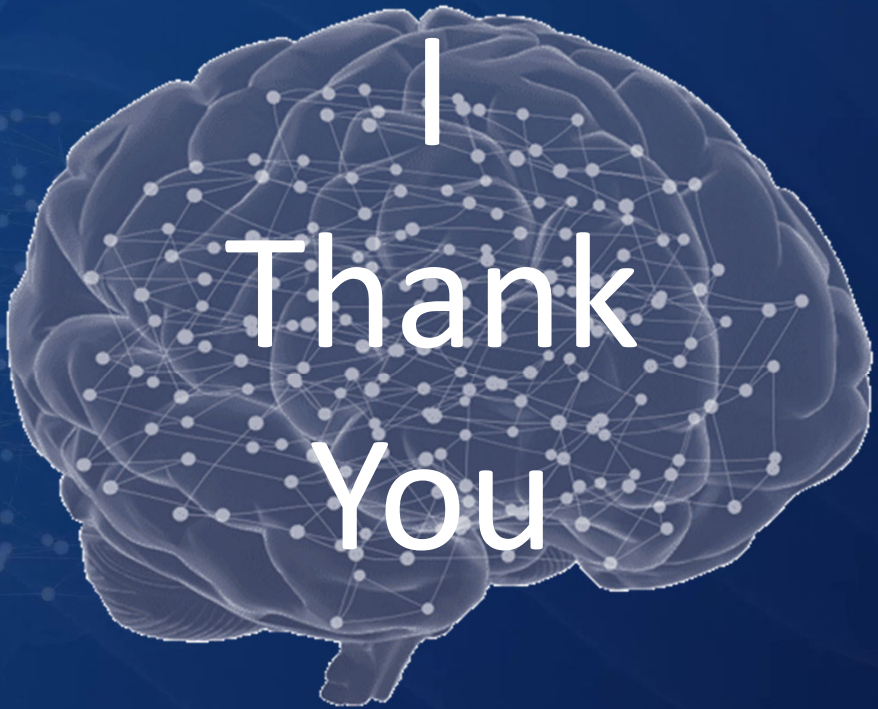


7 Conclusion

Motion Start-up Setter

7. Conclusion

- Mind is the central unit of the brain where thoughts are and thinking takes place.
- Understanding, reasoning, learning and initiating are the main tasks of the mind.
- Without logics, the mind could not work properly.
- The mind gives commands to act.
- The mind observes permanently its initiated commands for control and correction.



I
Thank
You